



Customer Shopping Behavior Analysis: Insights from Python, PostgreSQL, and Power BI



1 Enhancing marketing strategies via Python-driven customer segmentation and preference analysis

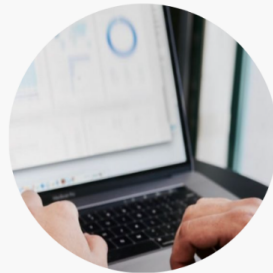


Understanding Customer Demographics



Age group analysis

Examining the distribution of customers across various age segments to tailor marketing efforts effectively and identify key demographics.



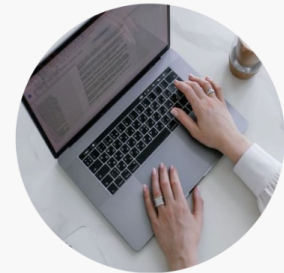
Gender distribution insights

Analyzing the proportion of male and female customers to understand their respective purchasing power and preferences for targeted campaigns.



Subscription status impact

Investigating how subscription status influences customer behavior and revenue generation, distinguishing between subscribers and non-subscribers.



Geographic customer segmentation

Segmenting customers by location to identify regional purchasing trends and optimize localized marketing strategies for better engagement.

Analyzing Purchasing Behavior



Product category popularity

Identifying the most frequently purchased product categories to inform inventory management and promotional strategies, maximizing sales potential.



Average purchase amount

Calculating the typical spending per transaction to understand customer value and identify opportunities for upselling or cross-selling initiatives.



High-spending customer identification

Pinpointing customers who consistently spend above average, especially those receiving discounts, for loyalty program enrollment and exclusive offers.



Payment method preferences

Understanding the most common payment methods used by customers to streamline checkout processes and enhance the overall shopping experience.

Optimizing Product Performance



Top-rated product analysis

Identifying products with the highest average customer review ratings to highlight successful items and inform future product development.



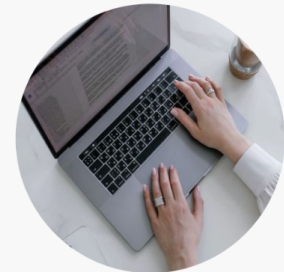
Customer review sentiment

Analyzing customer review ratings to gauge satisfaction levels and pinpoint areas for product improvement or service enhancement, boosting brand loyalty.



Product recommendation systems

Leveraging purchasing behavior data to develop personalized product recommendations, increasing customer engagement and driving additional sales.

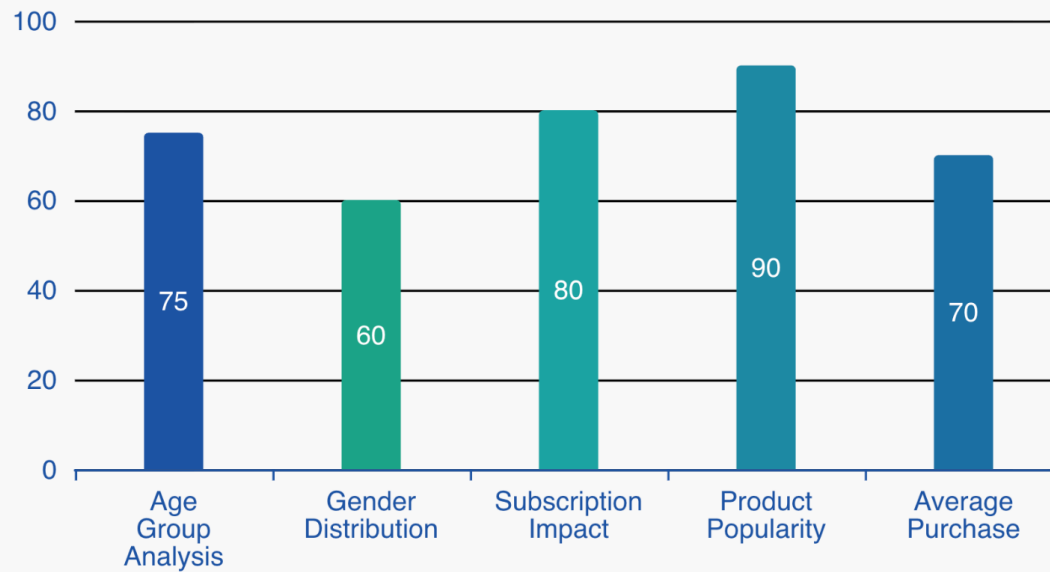


Impact of discounts

Assessing how discounts influence purchasing decisions and customer loyalty, optimizing promotional strategies for maximum effectiveness and revenue.



Customer Segmentation and Behavior Analysis



Key Customer Behavior Metrics(Effectiveness Score (%))

2 Harnessing PostgreSQL for profound SQL-driven business analysis and revenue insights.



Revenue Generation Insights

Male customers higher revenue

Analysis revealed male customers generated significantly higher total revenue compared to female customers, indicating a key demographic for sales focus.

Non-subscribers revenue contribution

Despite similar average purchase amounts, non-subscribers contributed more total revenue due to their larger overall customer base.



High-spending customer identification

Identified customers with above-average purchase amounts who also received discounts, presenting prime candidates for targeted loyalty programs.

Top-selling product categories

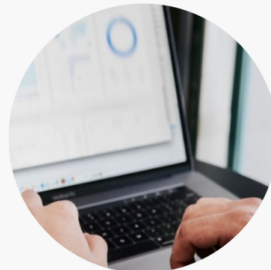
Clothing emerged as the highest-selling product category, suggesting a strong market demand and potential for further expansion.

Customer Behavior Analysis



Young Adult customer segment

Young Adult customers constituted the largest customer group, highlighting their importance in overall market strategy and product targeting.



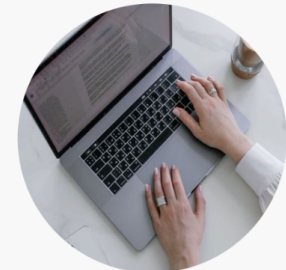
Average purchase amount

The average purchase amount was approximately £60, providing a benchmark for sales performance and customer spending habits.



Average customer review rating

An average customer review rating of 3.75 indicates generally positive customer satisfaction across the product range.



Subscription status impact

Most customers were non-subscribers, yet their collective spending significantly influenced total revenue figures.

Product Performance Metrics



Highest-rated products

Products such as Gloves, Sandals, and Boots received the highest average customer ratings, signifying strong customer satisfaction and product quality.



Product category sales

Clothing consistently ranked as the top-selling product category, underscoring its market dominance and consumer preference.



Review rating distribution

The distribution of review ratings provided insights into specific product strengths and areas potentially requiring improvement.

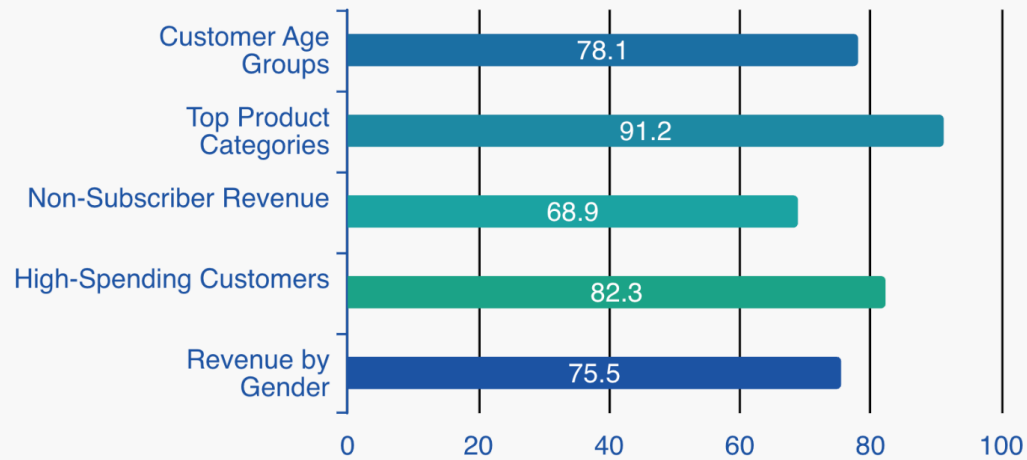


Customer satisfaction drivers

High ratings for certain products suggest specific features or qualities that resonate well with customers, driving satisfaction.



Revenue and Customer Behavior Analysis



Key Revenue and Customer Behavior Metrics(Impact Score (%))



Thank You

